Unit 9 · Algebraic Expressions

LEARNING GOAL

I can add and subtract linear expressions.

4

I can write and simplify an expression from a real-world situation.

LOOKS LIKE Mia earns $(3x + 5)$ dollars on Saturday and $(2.5x - 1)$ dollars on Sunday. Her total is $5.5x + 4$ dollars.

3

I can add and subtract linear expressions with rational coefficients.

LOOKS LIKE $\left(\frac{3}{4}x + 2\right) - \left(\frac{1}{4}x - 5\right) = \frac{1}{2}x + 7$

2

I can combine like terms with integer coefficients.

LOOKS LIKE $(3x + 5) + (2x - 1) = 5x + 4$

1

I need help to start this.

START HERE circle the x-terms, box the numbers, then combine each kind.

Level 3 is the goal for everyone: it means the benchmark is mastered.

LEARNING GOAL

I can tell whether two expressions are equivalent.

4 **I can justify equivalence using properties of operations, including factoring.**

LOOKS LIKE $12x + 18 = 6(2x + 3)$ by the distributive property: $6 \cdot 2x = 12x$ and $6 \cdot 3 = 18$.

3 **I can decide whether two linear expressions are equivalent and rewrite one as the other.**

LOOKS LIKE $2(3x - 1) + x$ and $7x - 2$: $6x - 2 + x = 7x - 2$, so they are equivalent.

2 **I can use the distributive property on a simple expression.**

LOOKS LIKE $3(2x + 4) = 6x + 12$

1 **I need help to start this.**

START HERE multiply the number outside by each term inside.

Level 3 is the goal for everyone: it means the benchmark is mastered.

LEARNING GOAL

I can write and solve two-step equations.

4

I can write a two-step equation from a real-world context, solve it, and check the answer.

LOOKS LIKE A gym charges $20 to join plus $15 a month. Ana paid $95: $15m + 20 = 95$, so $m = 5$. Check: $15(5) + 20 = 95$.

3

I can solve a two-step equation, including with rational numbers or the distributive property.

LOOKS LIKE $2(x + 1.5) = 9 \rightarrow x + 1.5 = 4.5 \rightarrow x = 3$

2

I can solve a one-step equation.

LOOKS LIKE $4x = -20 \rightarrow x = -5$

1

I need help to start this.

START HERE undo in reverse order. For $3x + 4 = 19$: subtract 4, then divide by 3.

Level 3 is the goal for everyone: it means the benchmark is mastered.

LEARNING GOAL

I can write and solve one-step inequalities and graph the solutions.

4

I can write an inequality from a real-world situation, solve it, and interpret the solution.

LOOKS LIKE Each video uses 0.75 GB and Ana has 4.5 GB left: $0.75v \leq 4.5$, so $v \leq 6$. She can watch at most 6 videos.

3

I can solve a one-step inequality, including with negative numbers, and graph the solution.

LOOKS LIKE $-3x < 12 \rightarrow x > -4$ (divide by –3 and flip)



2

I can graph an inequality on a number line.

LOOKS LIKE $x \geq -2$: closed circle at –2, shaded right



1

I need help to start this.

START HERE open circle means not included; closed circle means included.

Level 3 is the goal for everyone: it means the benchmark is mastered.